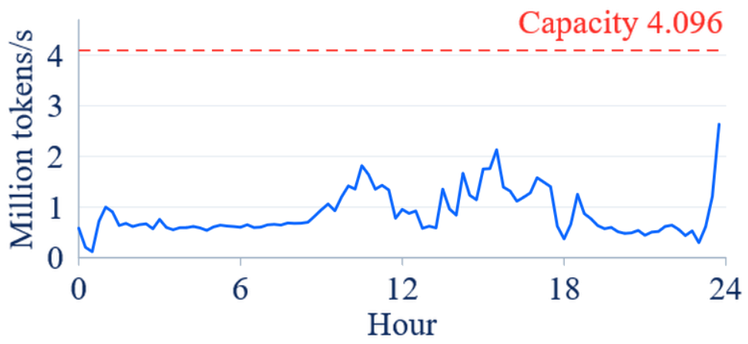
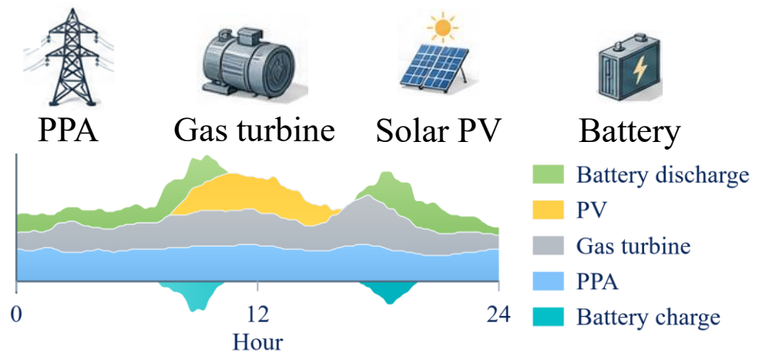
A Fixed predicted inputs

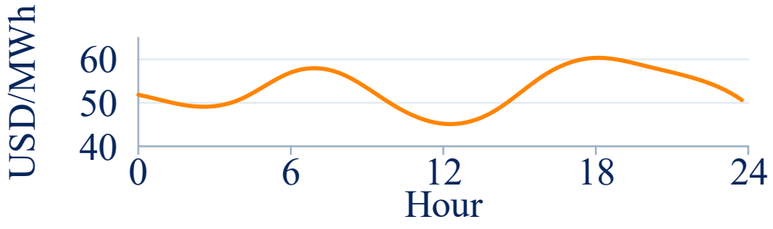
Workload arrivals (predicted)



Energy profile (fixed trajectories)



Day-ahead electricity price



B Joint optimization



Optimization objective

Profit = Base revenue – discounts – energy cost

Contract design

MILP master

- Discrete deferrals D_i
- Analytical discounts δ_i
- Monotonicity | Discount cap | Capacity

Service windows $a \le s \le \min(a + D_i, S)$

Benders' decomposition

Energy cost + dual cuts

Workload scheduling

LP subproblem

- Arrival-deadline windows
- Capacity constraint
- Complete all workload within the horizon

Facility demand ⟷ Residual grid imports

C Contract menu & evaluation

➢ Optimized contract menu

User type	Max deferral	Discount
Type 0	Immediate	0%
Types 1-3	0.25 h	3.40%
Types 4-5	0.75 h	8.38%
Type 6	1.00h	10.28%
Types 7-8	2.00 h	16.18%

Group workload schedules | Grid imports

➢ AI data center profit across power profiles

- Fix the LLM workload scenario

